United States Patent Application Publication	20250280498
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	KAMIOKA; Yuna et al.

INTERCONNECT SUBSTRATE AND METHOD OF MAKING INTERCONNECT SUBSTRATE

Abstract

An interconnect substrate includes a first interconnect layer having a first surface in which a recess is formed, an insulating layer that has a second surface facing the first surface and covers the first interconnect layer, a via hole overlapping the first interconnect layer in plan view and penetrating the insulating layer, and a second interconnect layer formed on the insulating layer and situated in contact with the first interconnect layer through the via hole.

Inventors: **KAMIOKA; Yuna (Nagano, JP), KODANI; Kotaro (Nagano, JP)**

Applicant: **SHINKO ELECTRIC INDUSTRIES CO., LTD.** (Nagano, JP)

Family ID: **96880832**

Appl. No.: **19/062616**

Filed: **February 25, 2025**

Foreign Application Priority Data

JP	2024-031435	Mar. 01, 2024

Publication Classification

Int. Cl.: **H05K1/11** (20060101); **H05K3/42** (20060101); **H05K3/46** (20060101)

U.S. Cl.:

CPC **H05K1/116** (20130101); H05K3/42 (20130101); H05K3/4644 (20130101); H05K2201/09563 (20130101); H05K2201/09827 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on and claims priority to Japanese Patent Application No. 2024-031435 filed on Mar. 1, 2024, with the Japanese Patent Office, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The disclosures herein relate to interconnect substrates and methods of making an interconnect substrate.

BACKGROUND

[0003] Manufacturing an interconnect substrate generally involves forming an insulating layer over an interconnect layer, creating a via hole in the insulating layer, performing a desmearing process, and forming another interconnect layer within the via hole and on the insulating layer.

[0004] In recent years, improving the connection reliability between interconnect layers has become a priority

[0005] It may be desired to provide an interconnect substrate structured to improve connection reliability between interconnect layers, and a method of making such an interconnect substrate.

RELATED ART DOCUMENTS

Patent Documents

[0006] [Patent Document 1] Japanese Laid-Open Patent Publication No. 2000-244127

SUMMARY

[0007] According to an aspect of the embodiment, an interconnect substrate includes a first interconnect layer having a first surface in which a recess is formed, an insulating layer that has a second surface facing the first surface and covers the first interconnect layer, a via hole overlapping the first interconnect layer in plan view and penetrating the insulating layer, and a second interconnect layer formed on the insulating layer and situated in contact with the first interconnect layer through the via hole, wherein in a cross section including a center line that is perpendicular to the second surface and passes through a center of an upper edge of the via hole in plan view, a surface of the recess includes a first point located at an edge of the recess and a second point located on the center line, and when a portion of the surface of the recess between the first point and the second point is approximated by a curve with only one inflection point, a first angle between a first line segment connecting the first point and a third point and a line segment extending from the first point to the center line on the second surface is less than 90 degrees, a second angle between the first line segment and a second line segment connecting the third point and a fourth point is greater than 90 degrees, a third angle between the second line segment and a third line segment connecting the fourth point and the second point is greater than 90 degrees, the third point is a point where a curvature of the curve is maximum between the first point and the inflection point, and the fourth point is a point where the curvature of the curve is maximum between the second point and the inflection point

[0008] The object and advantages of the embodiment will be realized and attained by means of the elements and combinations particularly pointed out in the claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention, as claimed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. **1** is a cross-sectional view illustrating an example of an interconnect substrate according to an embodiment;

[0010] FIG. **2** is a view illustrating an example of positional relationships mainly between a via

hole and a recess;
[0011] FIG. **3** is a cross-sectional view illustrating an example of the shape of the recess;
[0012] FIGS. **4**A through **4**C are cross-sectional views illustrating an example of a method of making the interconnect substrate according to the embodiment;
[0013] FIGS. **5**A through **5**C are cross-sectional views illustrating the example of the method of making the interconnect substrate according to the embodiment;
[0014] FIGS. **6**A and **6**B are cross-sectional views illustrating the example of the method of making the interconnect substrate according to the embodiment;
[0015] FIGS. **7**A through **7**C are cross-sectional views illustrating an example of a method of forming the via hole, the recess, and an interconnect layer;
[0016] FIGS. **8**A and **8**B are cross-sectional views illustrating the example of the method of forming the via hole, the recess, and the interconnect layer; and
[0017] FIGS. **9**A and **9**B are cross-sectional views illustrating the example of the method of forming the via hole, the recess, and the interconnect layer.

DESCRIPTION OF EMBODIMENTS

[0018] Improving connection reliability between interconnect layers may conceivably be achieved by forming a recess on the surface of the interconnect layer upon forming a via hole, and forming another interconnect layer by plating that fills the interior of the recess. However, upon manufacturing such an interconnect substrate, the inventors of the present invention found that the plating solution was not supplied to the edge of the recess, resulting in the formation of a gap in the recess. Any gap formed in the recess may pose a risk of failing to improve the connection reliability sufficiently. Upon this finding, the inventors of the present invention have made further study to prevent the occurrence of a gap in the recess, and have come up with the following embodiments.
[0019] In the following, an embodiment of the present disclosure will be described with reference to the accompanying drawings. In the present specification and the drawings, components having substantially the same functional configuration are referred to by the same reference numerals, and a duplicate description thereof may be omitted.

Structure of Interconnect Substrate of Embodiment

[0020] First, the structure of an interconnect substrate according to an embodiment will be described. FIG. **1** is a cross-sectional view illustrating an example of an interconnect substrate according to an embodiment.
[0021] As illustrated in FIG. **1**, an interconnect substrate **1** according to the embodiment includes, for example, a core layer **100**, a buildup layer **200** formed on the A-side surface of the core layer **100**, and a buildup layer **300** formed on the B-side surface of the core layer **100**. The interconnect substrate **1** may be a coreless substrate without a core layer.
[0022] In this embodiment, for convenience, the side of an object oriented in the same direction as the buildup layer **200** side of the core layer **100** is referred to as an upper side or an A side, and the side of the object oriented in the same direction as the buildup layer **300** side of the core layer **100** is referred to as a lower side or a B side. The surface of an object on the upper side thereof is referred to as an A-side surface or an upper surface, and the surface of the object on the lower side thereof is referred to as a B-side surface or the lower surface. However, the interconnect substrate **1** may be placed upside down when used, or may be arranged at any angle. The plan view refers to the view of an object as seen from the direction normal to the A-side surface of the core layer **100**. The plane shape refers to the shape of an object as viewed from the direction normal to the A-side surface of the core layer **100**.
[0023] The core layer **100** includes an insulating substrate **111** having a plurality of through holes **112** formed therethrough and an electrically conductive layer **113** formed on the inner surfaces of the through holes **112**. The material of the substrate is, for example, glass epoxy or the like, and the material of the conductive layer **113** is, for example, copper (Cu) or the like. The core layer **100** may have a filling material filling the space inside the conductive layer **113**.

[0024] The buildup layer **200** includes interconnect layers **210**, **230** and **250**, insulating layers **410** and **420**, and a solder resist layer **260**. The buildup layer **300** includes interconnect layers **310**, **330** and **350**, insulating layers **510** and **520**, and a solder resist layer **360**. The insulating layer **410** is disposed between the interconnect layers **210** and **230** adjacent to each other in the thickness direction. The insulating layer **420** is disposed between the interconnect layers **230** and **250** adjacent to each other in the thickness direction. The insulating layer **510** is disposed between the interconnect layers **310** and **330** adjacent to each other in the thickness direction. The insulating layer **520** is disposed between the interconnect layers **330** and **350** adjacent to each other in the thickness direction. The insulating layer **410** includes an insulating layer **220** and an adhesion enhancing film **270**, and the insulating layer **420** includes an insulating layer **240** and an adhesion enhancing film **280**. The insulating layer **510** includes an insulating layer **320** and an adhesion enhancing film **370**, and the insulating layer **520** includes an insulating layer **340** and an adhesion enhancing film **380**.
[0025] The interconnect layer **210** is formed on the A-side surface of the core layer **100**. The interconnect layer **310** is formed on the B-side surface of the core layer **100**. The interconnect layers **210** and **310** are electrically connected to each other by the conductive layer **113**. The material of the interconnect layers **210** and **310** is, for example, copper (Cu) or the like. The thicknesses of the interconnect layers **210** and **310** are, for example, about 10 μm to 30 μm.
[0026] The adhesion enhancing film **270** is formed on the A-side surface and the side surfaces of the interconnect layer **210**. The adhesion enhancing film **270** covers the A-side surface and the side surfaces of the interconnect layer **210**. The adhesion enhancing film **270** is formed of a material having two types of functional groups with different reactivities within a single molecule. The material of the adhesion enhancing film **270** is, for example, a silane coupling agent or a titanium coupling agent. The thickness of the adhesion enhancing film **270** is, for example, about 3 nm to 8 nm.
[0027] In the silane coupling agent, a functional group which chemically bonds with an organic material such as a resin preferably contains an amino group, an epoxy group, a mercapto group, an isocineate group, a methacryloxy group, an acryloxy group, a ureido group, or a sulfide group. An optimum functional group is selected according to the type of resin to be chemically bonded with the silane coupling agent.
[0028] In the silane coupling agent, a functional group which chemically bonds with an inorganic material such as a metal preferably contains an azole group, a silanol group, a methoxy group, or an ethoxy group. An optimum functional group is selected according to the type of metal to be chemically bonded with the silane coupling agent.
[0029] The insulating layer **220** is formed so as to cover the interconnect layer **210** and the adhesion enhancing film **270** on the A-side surface of the core layer **100**. The material of the insulating layer **220** is, for example, an insulating resin mainly composed of an epoxy-based resin or a polyimide-based resin. The thickness of the insulating layer **220** is, for example, about 30 μm to 40 μm. The insulating layer **220** may contain a filler such as silica (SiO.sub.2). The filler content in the insulating layer **220** may be appropriately set according to the required coefficient of thermal expansion (CTE). The adhesion enhancing film **270** improves adhesion between the interconnect layer **210** and the insulating layer **220**. That is, the provision of the adhesion enhancing film **270** achieves higher adhesion between the interconnect layer **210** and the insulating layer **220** than when the interconnect layer **210** and the insulating layer **220** are in direct contact.
[0030] A recess **215** is formed on the A-side surface of the interconnect layer **210**. A via hole **412** is formed in the insulating layer **410**. The via hole **412** penetrates the insulating layer **410**. That is, the via hole **412** penetrates the insulating layer **220** and the adhesion enhancing film **270**. The via hole **412** overlaps the interconnect layer **210** in plan view and reaches the interconnect layer **210**. The via hole **412** overlaps the recess **215** in plan view and connects to the recess **215**.
[0031] The interconnect layer **230** is formed on the A-side surface of the insulating layer **220**. The

interconnect layer **230** includes a via conductor positioned within the via hole **412** and the recess **215**, and an interconnect pattern on the A-side surface of the insulating layer **220**. The interconnect pattern of the interconnect layer **230** is electrically connected to the interconnect layer **210** via the via conductor. The material and thickness of the interconnect layer **230** are substantially the same as, for example, the interconnect layer **210**.

[0032] The adhesion enhancing film **280** is formed on the A-side surface and the side surfaces of the interconnect layer **230**. The adhesion enhancing film **280** covers the A-side surface and the side surfaces of the interconnect layer **230**. The material and thickness of the adhesion enhancing film **280** are substantially the same as, for example, the adhesion enhancing film **270**.

[0033] The insulating layer **240** is formed so as to cover the interconnect layer **230** and the adhesion enhancing film **280** on the A-side surface of the insulating layer **220**. The material and thickness of the insulating layer **240** are, for example, substantially the same as those of the insulating layer **220**. The insulating layer **240** may contain a filler such as silica (SiO.sub.2). The content of the filler in the insulating layer **240** is, for example, substantially the same as that of the insulating layer **220**. The adhesion enhancing film **280** improves adhesion between the interconnect layer **230** and the insulating layer **240**. That is, the provision of the adhesion enhancing film **280** achieves higher adhesion between the interconnect layer **230** and the insulating layer **240** than when the interconnect layer **230** and the insulating layer **240** are in direct contact.

[0034] A recess **235** is formed on the A-side surface of the interconnect layer **230**. A via hole **422** is formed in the insulating layer **420**. The via hole **422** penetrates the insulating layer **420**. That is, the via hole **422** penetrates the insulating layer **240** and the adhesion enhancing film **280**. The via hole **422** overlaps the interconnect layer **230** in plan view and reaches the interconnect layer **230**. The via hole **422** overlaps the recess **235** in plan view and connects to the recess **235**.

[0035] The interconnect layer **250** is formed on the A-side surface of the insulating layer **240**. The interconnect layer **250** includes a via conductor positioned within the via hole **422** and the recess **235**, and an interconnect pattern on the A-side surface of the insulating layer **240**. The interconnect pattern of the interconnect layer **250** is electrically connected to the interconnect layer **230** via the via conductor. The material and the thickness of the interconnect layer **250** are substantially the same as those of the interconnect layer **210**, for example. An adhesion enhancing film (not illustrated) may be formed on the A-side surface and the side surfaces of the interconnect layer **250**.

[0036] The solder resist layer **260** is formed on the A-side surface of the insulating layer **240** so as to cover the interconnect layer **250**. An opening **261** is formed in the solder resist layer **260**. The opening **261** penetrates the solder resist layer **260**. The opening **261** overlaps an electrode pad which is a part of the interconnect layer **250** in plan view and reaches the electrode pad.

[0037] The adhesion enhancing film **370** is formed on the B-side surface and the side surfaces of the interconnect layer **310**. The adhesion enhancing film **370** covers the B-side surface and the side surfaces of the interconnect layer **310**. The material and thickness of the adhesion enhancing film **370** are, for example, substantially the same as those of the adhesion enhancing film **270**.

[0038] The insulating layer **320** is formed on the B-side surface of the core layer **100** so as to cover the interconnect layer **310** and the adhesion enhancing film **370**. The material and thickness of the insulating layer **320** are, for example, substantially the same as those of the insulating layer **220**. The insulating layer **320** may contain a filler such as silica (SiO.sub.2). The filler content of the insulating layer **320** is, for example, substantially the same as that of the insulating layer **220**. The adhesion enhancing film **370** improves adhesion between the interconnect layer **310** and the insulating layer **320**. That is, the provision of the adhesion enhancing film **370** achieves higher adhesion between the interconnect layer **310** and the insulating layer **320** than when the interconnect layer **310** and the insulating layer **320** are in direct contact.

[0039] A recess **315** is formed on the B-side surface of the interconnect layer **310**. A via hole **512** is formed in the insulating layer **510**. The via hole **512** penetrates the insulating layer **510**. That is, the via hole **512** penetrates the insulating layer **320** and the adhesion enhancing film **370**. The via hole

512 overlaps the interconnect layer **310** in plan view and reaches the interconnect layer **310**. The via hole **512** overlaps the recess **315** in plan view and connects to the recess **315**.

[0040] The interconnect layer **330** is formed on the B-side surface of the insulating layer **320**. The interconnect layer **330** includes a via conductor positioned within the via hole **512** and the recess **315**, and an interconnect pattern on the B-side surface of the insulating layer **320**. The interconnect pattern of the interconnect layer **330** is electrically connected to the interconnect layer **310** via the via conductor. The material and the thickness of the interconnect layer **330** are substantially the same as those of the interconnect layer **210**, for example.

[0041] The adhesion enhancing film **380** is formed on the B-side surface and the side surfaces of the interconnect layer **330**. The adhesion enhancing film **380** covers the B-side surface and the side surfaces of the interconnect layer **330**. The material and thickness of the adhesion enhancing film **380** are, for example, substantially the same as those of the adhesion enhancing film **270**.

[0042] The insulating layer **340** is formed so as to cover the interconnect layer **330** and the adhesion enhancing film **380** on the B-side surface of the insulating layer **320**. The material and thickness of the insulating layer **340** are, for example, substantially the same as those of the insulating layer **220**. The insulating layer **340** may contain a filler such as silica (SiO.sub.2). The content of the filler in the insulating layer **340** is, for example, substantially the same as that of the insulating layer **220**. The adhesion enhancing film **380** improves the adhesion between the interconnect layer **330** and the insulating layer **340**. That is, when the adhesion enhancing film **380** is provided, the adhesion between the interconnect layer **330** and the insulating layer **340** is higher than when the interconnect layer **330** and the insulating layer **340** are in direct contact.

[0043] A recess **335** is formed on the B-side surface of the interconnect layer **330**. A via hole **522** is formed in the insulating layer **520**. The via hole **522** penetrates the insulating layer **520**. That is, the via hole **522** penetrates the insulating layer **340** and the adhesion enhancing film **380**. The via hole **522** overlaps the interconnect layer **330** in plan view and reaches the interconnect layer **330**. The via hole **522** overlaps the recess **335** in plan view and connects to the recess **335**.

[0044] The interconnect layer **350** is formed on the B-side surface of the insulating layer **340**. The interconnect layer **350** includes a via conductor situated within the via hole **522** and the recess **335**, and an interconnect pattern on the B-side surface of the insulating layer **340**. The interconnect pattern of the interconnect layer **350** is electrically connected to the interconnect layer **330** via the via conductor. The material and thickness of the interconnect layer **350** are substantially the same as, for example, the interconnect layer **210**. An adhesion enhancing film (not illustrated) may be formed on the B-side surface and the side surfaces of the interconnect layer **350**.

[0045] The solder resist layer **360** is formed on the B-side surface of the insulating layer **340** so as to cover the interconnect layer **350**. An opening **361** is formed in the solder resist layer **360**. The opening **361** penetrates the solder resist layer **360**. The opening **361** overlaps an electrode pad which is a part of the interconnect layer **350** in plan view and reaches the electrode pad.

[0046] The interconnect layer **210**, the adhesion enhancing film **270**, the insulating layer **220**, and the interconnect layer **230** will now be described in more detail. FIG. **2** is a view illustrating an example of positional relationships mainly between the via hole **412** and the recess **215**. FIG. **3** is a cross-sectional view illustrating an example of the shape of the recess **215**. FIG. **3** corresponds to a cross-sectional view along the line III-III in FIG. **2**.

[0047] As illustrated in FIGS. **2** and **3**, the recess **215** is formed on the upper surface **211**, which is the A-side surface of the interconnect layer **210**. The insulating layer **410** has a lower surface **411**, which is the B-side surface. The lower surface **411** of the insulating layer **410** faces the upper surface **211** of the interconnect layer **210**. The upper surface **211** of the interconnect layer **210** is an example of a first surface, and the lower surface **411** of the insulating layer **410** is an example of a second surface.

[0048] The via hole **412** has a hole portion **222** formed through the insulating layer **220** and a hole portion **272** formed through the adhesion enhancing film **270**. The hole portion **222** may be an inverted truncated conical recess. That is, the hole portion **222** is structured such that the opening diameter at the upper end may be larger than the opening diameter at the lower end. The hole portion **272** connects to the hole portion **222** and the recess **215**. The aperture of the hole portion **272** is larger than the aperture at the lower end of the hole portion **222**. The aperture at the upper end of the recess **215** is larger than the aperture of the hole portion **272**. In plan view, thus, the hole portion **272** is inside the upper end of the recess **215**, and the lower end of the hole portion **222** is inside the hole portion **272**.

[0049] As will be described later, the recess **215** may be formed by wet etching. The surface of the recess **215**, which is a part of the upper surface **211**, may have fine surface irregularities due to wet etching. In FIG. **3**, the fine irregularities of the surface of the recess **215** are omitted. FIG. **3** illustrates a cross section including a center line C**1** that is perpendicular to the lower surface **411** and passes through the center of the upper edge **413** of the via hole **412** in plan view. In this cross section, the surface of the recess **215** includes a first point **11** located on the edge of the recess **215** and a second point **12** located on the center line C**1**.

[0050] When the portion of the surface of the recess **215** between the first point **11** and the second point **12** is approximated by a curve with only one inflection point, the recess **215** includes a third point **13** having the maximum curvature between the first point **11** and an inflection point **19**, and a fourth point **14** having the maximum curvature between the second point **12** and the inflection point **19**.

[0051] The first angle between a first line segment **21** connecting the first point **11** and the third point **13** and a line segment extending from the first point to the center line CL on the lower surface **411** of the insulating layer **410** is acute, the second angle between the first line segment **21** and a second line segment **22** connecting the third point **13** and the fourth point **14** is obtuse, and the third angle between the second line segment **22** and a third line segment **23** connecting the fourth point **14** and the second point **12** is obtuse. In other words, the angle θ**1** representing the first angle is less than 90 degrees, the angle θ**2** representing the second angle is greater than 90 degrees, and the angle θ**3** representing the third angle is greater than 90 degrees.

[0052] As described above, the surface of the recess **215** has, on both sides of the center line C**1** in the cross-sectional view, a first inclined portion which gently slopes downwards from the first point **11** to the second point **12**, a second inclined portion which steeply slopes downwards from the first inclined portion, and a third inclined portion which gently slopes downwards from the second inclined portion. Generally, the first inclined portion corresponds to the portion between the first point **11** and the third point **13**, the second inclined portion corresponds to the portion between the third point **13** and the fourth point **14**, and the third inclined portion corresponds to the portion between the fourth point **14** and the second point **12**.

[0053] The interconnect layer **230** is in contact with the surface of the recess **215**, the portion of the lower surface **411** of the insulating layer **410** facing the recess **215**, the inner wall surface of the via hole **412**, and the A-side surface of the insulating layer **410**. The interconnect layer **230** has a seed layer **236** and a plating layer **237**. The seed layer **236** directly covers the surface of the recess **215**, the portion of the lower surface **411** of the insulating layer **410** facing the recess **215**, the inner wall surface of the via hole **412**, and the A-side surface of the insulating layer **410**. The plating layer **237** is provided on the seed layer **236**. The material of the seed layer **236** and the plating layer **237** is copper (Cu) or the like.

[0054] The insulating layer **410** has a first region **31** extending to a depth of 2 μm or less from the portion of the lower surface **411** facing the recess **215**. The first region **31** includes a second region **32** including the inner wall surface of the via hole, and a third region **33** connected to the second region **32** and located between the second region **32** and the first point **11** in plan view. The second region **32** and the third region **33** include the insulating layer **220**. The third region **33** includes the adhesion enhancing film **270**, but the second region **32** does not include the adhesion enhancing film **270**. As a result, when the insulating layer **220** includes a filler and the adhesion enhancing film **270** does not include the filler, the density of the filler in the second region **32** is higher than that in the third region **33**.

[0055] The interconnect layer **230**, the adhesion enhancing film **280**, the insulating layer **240**, and the interconnect layer **250** are also laminated to each other in substantially the same manner as the interconnect layer **210**, the adhesion enhancing film **270**, the insulating layer **220**, and the interconnect layer **230**. The interconnect layer **310**, the adhesion enhancing film **370**, the insulating layer **320**, and the interconnect layer **330** are also laminated to each other in substantially the same manner as the interconnect layer **210**, the adhesion enhancing film **270**, the insulating layer **220**, and the interconnect layer **230**. The interconnect layer **330**, the adhesion enhancing film **380**, the insulating layer **340**, and the interconnect layer **350** are also laminated to each other in substantially the same as manner the interconnect layer **210**, the adhesion enhancing film **270**, the insulating layer **220**, and the interconnect layer **230**.

[0056] In the relationship between the interconnect layers **210** and **230**, the interconnect layer **210** is an example of a first interconnect layer, and the interconnect layer **230** is an example of a second interconnect layer. In the relationship between the interconnect layers **230** and **250**, the interconnect layer **230** is an example of a first interconnect layer, and the interconnect layer **250** is an example of a second interconnect layer. In the relationship between the interconnect layers **310** and **330**, the interconnect layer **310** is an example of a first interconnect layer, and the interconnect layer **330** is an example of a second interconnect layer. In the relationship between the interconnect layers **330** and **350**, the interconnect layer **330** is an example of a first interconnect layer, and the interconnect layer **350** is an example of a second interconnect layer.

Method of Making Interconnect Substrate According to Embodiment

[0057] In the following, a method of making the interconnect substrate **1** according to the embodiment will be described. FIGS. **4**A through **4**C to FIGS. **6**A and **6**B are cross-sectional views illustrating an example of the method of making the interconnect substrate according to the embodiment.

[0058] First, as illustrated in FIG. **4**A, a laminate of the core layer **100**, the interconnect layer **210**, and the interconnect layer **310** is prepared. The core layer **100** includes the insulating substrate **111** having the through holes **112** and the conductive layer **113**.

[0059] Next, as illustrated in FIG. **4**B, the adhesion enhancing film **270** is formed on the upper and side surfaces of the interconnect layer **210**, and the adhesion enhancing film **370** is formed on the lower and side surfaces of the interconnect layer **310**. The adhesion enhancing films **270** and **370** are formed using, for example, a silane coupling agent.

[0060] To form the adhesion enhancing films **270** and **370** using the silane coupling agent, for example, the structure illustrated in FIG. **4**A may be immersed in a diluted solution of the silane coupling agent. Alternatively, the adhesion enhancing film **270** may be formed by spray-coating the upper and side surfaces of the interconnect layer **210** of the structure illustrated in FIG. **4**A with a diluted solution of the silane coupling agent, and the adhesion enhancing film **370** may be formed by spray-coating the lower and side surfaces of the interconnect layer **310** with a diluted solution of the silane coupling agent. The concentration of the diluted solution of the silane coupling agent is 0.1% to 10%, preferably 0.5% to 5%. At this stage, the thickest part of the adhesion enhancing films **270** and **370** may be about 20 nm to 30 nm, for example. Thereafter, washing with water, partial removal of the adhesion enhancing films **270** and **370** using a removal treatment solution such as an acidic solution, and drying are performed in this order. As a result, the adhesion enhancing films **270** and **370** having a thickness of about 3 nm to 8 nm are obtained.

[0061] After the formation of the adhesion enhancing films **270** and **370**, as illustrated in FIG. **4**C, an uncured resin film is attached to the A-side surface of the core layer **100** so as to cover the interconnect layer **210** and the adhesion enhancing film **270**, and an uncured resin film is attached to the B-side surface of the core layer **100** so as to cover the interconnect layer **310** and the adhesion enhancing film **370**. These resin films are cured by heat to form the insulating layers **220** and **320**. The insulating layers **220** and **320** are formed of an insulating resin such as epoxy resin or polyimide resin. The insulating **220** layers and **320** may alternatively be formed by applying a liquid resin. The insulating layers **220** and **320** are each an example of a first insulating layer.

[0062] As illustrated in FIG. **5**A, the via hole **412** reaching the interconnect layer **210** is formed in the insulating layer **410** (i.e., the insulating layer **220** and the adhesion enhancing film **270**), and the via hole **512** reaching the interconnect layer **310** is formed in the insulating layer **510** (i.e., the insulating layer **320** and the adhesion enhancing film **370**). In addition, the recess **215** is formed in the A-side surface of the interconnect layer **210**, and the recess **315** is formed on the B-side surface of the interconnect layer **310**.

[0063] Subsequently, as illustrated in FIG. **5**B, the interconnect layer **230** including a via conductor positioned within the via hole **412** and the recess **215** and an interconnect pattern on the A-side surface of the insulating layer **220** is formed. The interconnect layer **330** including a via conductor positioned within the via hole **512** and the recess **315** and an interconnect pattern on the B-side surface of the insulating layer **320** is also formed.

[0064] A method of forming the via hole **412**, the recess **215**, and the interconnect layer **230** will now be described. FIGS. **7**A through **7**C to FIGS. **9**A and **9**B are cross-sectional views illustrating an example of a method of forming via holes, recesses and interconnect layers.

[0065] First, as illustrated in FIG. **7**A, the insulating layer **410** (i.e., the insulating layer **220** and the adhesion enhancing film **270**) is machined with a laser to form the hole portion **222** in the insulating layer **220** and a hole portion **272**X in the adhesion enhancing film **270**. The hole portion **222** penetrates the insulating layer **220**, and the hole portion **272**X penetrates the adhesion enhancing film **270**. The hole portions **222** and **272**X are formed so as to overlap the interconnect layer **210** in plan view. The insulating layer **220** and the adhesion enhancing film **270** may alternatively be machined by a drill. The hole portion **222** is an example of a first hole portion.

[0066] Next, desmearing is performed. As a result of the desmearing, as illustrated in FIG. **7**B, the portion of the adhesion enhancing film **270** exposed to the hole portion **272**X is wet etched through the hole portion **222**, which results in the formation of the hole portion **272** in the adhesion enhancing film **270** which is wider than the lower end of the hole portion **222**. The hole portion with the **272** overlaps interconnect layer **210** in plan view. These steps achieve the formation of the via hole **412** including the hole portions **222** and **272** in the insulating layer **410**. The interconnect layer **210** is exposed through the via hole **412**. For desmearing, for example, sodium permanganate solution is used. The hole portion **272** is an example of a second opening.

[0067] It may be noted that it is not necessary to form the hole portion **272**X in the adhesion enhancing film **270** when machining with a laser or a drill. Even without the presence of the hole portion **272**X, the desmearing through the hole portion **222** is able to form the hole portion **272**.

[0068] As illustrated in FIG. **7**C, the interconnect layer **210** is wet etched through the hole portions **222** and **272** to form the recess **215** in the upper surface **211** of the interconnect layer **210**. For example, sodium persulfate is used for the wet etching of the interconnect layer **210**.

[0069] During the formation of the recess **215**, the presence of the hole portion **272** in the adhesion enhancing film **270** causes the interconnect layer **210** to be wet etched in a larger area from the beginning than when the hole portion **272** is not provided. As a result, wet etching progresses more strongly in the thickness direction. Further, in the portion where the adhesion enhancing film **270** remains, the etchant is less likely to penetrate into the interface between the interconnect layer **210** and the adhesion enhancing film **270** than the interface between the interconnect layer **210** and the insulating layer **220** which would exist if the adhesion enhancing film **270** were not provided. That is, wet etching in the direction parallel to the upper surface of the interconnect layer **210** is less likely to occur. Therefore, the recess **215** having the illustrated cross-sectional shape is formed.

[0070] Subsequently, as illustrated in FIG. **8**A, the seed layer **236** made of copper or the like is formed so as to directly cover the surface of the recess **215**, the portion of the lower surface **411** of the insulating layer **410** exposed to the recess **215**, the inner wall surface of the via hole **412**, and the A-side surface of the insulating layer **410**. The seed layer **236** may be formed by, for example, sputtering or electroless plating.

[0071] As illustrated in FIG. **8**B, a plating resist layer **238** is formed on the seed layer **236**. The plating resist layer **238** is exposed and developed to form an opening **239** in the plating resist layer **238**.

[0072] As illustrated in FIG. **9**A, the plating layer **237** made of copper or the like is formed in the opening **239** of the plating resist layer **238** by an electrolytic plating method, using the seed layer **236** as a power supply path for plating.

[0073] Subsequently, as illustrated in FIG. **9**B, the plating resist layer **238** is removed. Further, with the plating layer **237** being used as a mask, the seed layer **236** is removed by flash etching. Through these steps, the interconnect layer **230** including the seed layer **236** and the plating layer **237** is effectively formed.

[0074] The via hole **512** and the interconnect layer **330** may also be formed in substantially the same manner as the via hole **412** and the interconnect layer **230**.

[0075] After the interconnect layers **230** and **330** are formed, as illustrated in FIG. **5**C, the adhesion enhancing film **280** is formed on the upper and side surfaces of the interconnect layer **230**, and the adhesion enhancing film **380** is formed on the lower and side surfaces of the interconnect layer **330**. The adhesion enhancing films **280** and **380** may be formed in substantially the same manner as the adhesion enhancing films **270** and **370**. Further, the insulating layer **240** is formed on the insulating layer **220** so as to cover the interconnect layer **230** and the adhesion enhancing film **280**. The insulating layer **340** is also formed under the insulating layer **320** so as to cover the interconnect layer **330** and the adhesion enhancing film **380**. The insulating layers **240** and **340** may be formed in substantially the same manner as the insulating layers **220** and **320**. The insulating layers **240** and **340** are each an example of a first insulating layer.

[0076] As illustrated in FIG. **6**A, the via hole **422** reaching the interconnect layer **230** is formed in the insulating layer **420** (i.e., the insulating layer **240** and the adhesion enhancing film **280**), and the via hole **522** reaching the interconnect layer **330** is formed in the insulating layer **520** (i.e., the insulating layer **340** and the adhesion enhancing film **380**). Further, the recess **235** is formed on the A-side surface of the interconnect layer **230**, and the recess **335** is formed on the B-side surface of the interconnect layer **330**. Subsequently, the interconnect layer **250** including a via conductor positioned within the via hole **422** and the recess **235** and an interconnect pattern on the A-side surface of the insulating layer **240** is formed. Also, the interconnect layer **350** including a via conductor located within the via hole **522** and the recess **335** and an interconnect pattern on the B-side surface of the insulating layer **340** is formed.

[0077] The via hole **422**, the recess **235**, and the interconnect layer **250** may be formed in substantially the same manner as the via hole **412**, the recess **215**, and the interconnect layer **230**. The via hole **522**, the recess **335**, and the interconnect layer **350** may be formed in substantially the same manner as the via hole **412**, the recess **215**, and the interconnect layer **230**.

[0078] As illustrated in FIG. **6 6**B, the solder resist layer **260** is formed on the insulating layer **240** and the solder resist layer **360** is formed under the insulating layer **340**. Thereafter, the opening **261** is formed in the solder resist layer **260** and the opening **361** is formed in the solder resist layer **360**.

[0079] By following these steps, the successful manufacture of the interconnect substrate **1** according to the embodiment is achieved.

[0080] In the interconnect substrate **1**, when the portion between the first point **11** and the second point **12** on the surface of the recess **215** is approximated by a curve with only one inflection point **19**, the first angle θ**1** is less than 90 degrees, the second angle θ**2** is greater than 90 degrees, and the third angle θ**3** is greater than 90 degrees. As a result, during the formation of the plating layer **237**, the plating liquid easily wets and spreads to the edge of the recess **215**, so that the recess **215** is easily filled with the interconnect layer **230** having the seed layer **236** and the plating layer **237**. That is, a gap is unlikely to remain in the recess **215**. The reliability of connection between the

interconnect layer **210** and the interconnect layer **230** is thus effectively improved.
[0081] In particular, when the angle θ**1** is from 10 degrees to 40 degrees, the angle θ**2** is from 120 degrees to 160 degrees, and the angle θ**3** is from 120 degrees to 160 degrees, the plating liquid easily wets and spreads to the edge of the recess **215** during the formation of the plating layer **237**.
[0082] In the interconnect substrate **1**, it may sometimes be difficult to distinguish between the adhesion enhancing film **270** and the insulating layer **220**. Even in such a case, the presence of the adhesion enhancing film **270** is effectively estimated by the presence of the second region **32** and the third region **33** having different filler densities.
[0083] The plane shape of the via hole does not have to be a circle. When the plane shape of the via hole is not a circle, the center of the circle obtained by approximating the plane shape of the via hole may be set as the center in plan view.
[0084] The surface of the recess **215** may be roughened between the formation of the recess **215** by wet etching and the formation of the seed layer **236**.
[0085] According to the disclosed technique, the reliability of connections between interconnect layers is effectively improved.
[0086] The present disclosures non-exhaustively include the subject matter set out in the following clauses.
[0087] Clause 1. A method of making an interconnect substrate, comprising: [0088] forming a first interconnect layer having a first surface; [0089] forming an adhesion enhancing film covering the first interconnect layer; [0090] forming a first insulating layer on the adhesion enhancing film; [0091] forming a first hole portion that penetrates the first insulating layer, and overlaps the first interconnect layer in plan view; [0092] wet etching the adhesion enhancing film through the first hole portion to form a second hole portion that penetrates the adhesion enhancing film and is wider than a lower end of the first hole portion; [0093] wet etching the first interconnect layer through the first hole portion and the second hole portion to form a recess in the first surface; and [0094] forming a second interconnect layer on the first insulating layer in contact with the first interconnect layer through the first hole portion and the second hole portion.
[0095] Clause The 2. method of making an interconnect substrate as recited in clause 1, wherein the adhesion enhancing film includes a silane coupling agent.
[0096] Clause 3. The method of making an interconnect substrate as recited in clause 1, wherein the wet etching the first interconnect layer includes using an acidic solution.
[0097] Clause 4. The method of making an interconnect substrate as recited in clause 1, wherein the wet etching the adhesion enhancing film includes using a sodium permanganate solution.
[0098] Clause 5. The method of making an interconnect substrate as recited in clause 1, wherein an insulating layer which is a laminate of the adhesion enhancing film and the first insulating layer has a second surface facing the first surface, [0099] wherein, in a cross section including a center line that is perpendicular to the second surface and passes through a center of an upper edge of the first hole portion in plan view, [0100] a surface of the recess includes a first point located at an edge of the recess and a second point located on the center line, and [0101] when a portion of the surface of the recess between the first point and the second point is approximated by a curve with only one inflection point, [0102] a first angle between a first line segment connecting the first point and a third point and a line segment extending from the first point to the center line on the second surface is less than 90 degrees, [0103] a second angle between the first line segment and a second line segment connecting the third point and a fourth point is greater than 90 degrees, [0104] a third angle between the second line segment and a third line segment connecting the fourth point and the second point is greater than 90 degrees, [0105] the third point is a point where a curvature of the curve is maximum between the first point and the inflection point, and [0106] the fourth point is a point where the curvature of the curve is maximum between the second point and the inflection point.
[0107] All examples and conditional language recited herein are intended for pedagogical purposes

to aid the reader in understanding the invention and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiment(s) of the present inventions have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. An interconnect substrate comprising: a first interconnect layer having a first surface in which a recess is formed; an insulating layer that has a second surface facing the first surface and covers the first interconnect layer; a via hole overlapping the first interconnect layer in plan view and penetrating the insulating layer; and a second interconnect layer formed on the insulating layer and situated in contact with the first interconnect layer through the via hole, wherein in a cross section including a center line that is perpendicular to the second surface and passes through a center of an upper edge of the via hole in plan view, a surface of the recess includes a first point located at an edge of the recess and a second point located on the center line, and when a portion of the surface of the recess between the first point and the second point is approximated by a curve with only one inflection point, a first angle between a first line segment connecting the first point and a third point and a line segment extending from the first point to the center line on the second surface is less than 90 degrees, a second angle between the first line segment and a second line segment connecting the third point and a fourth point is greater than 90 degrees, a third angle between the second line segment and a third line segment connecting the fourth point and the second point is greater than 90 degrees, the third point is a point where a curvature of the curve is maximum between the first point and the inflection point, and the fourth point is a point where the curvature of the curve is maximum between the second point and the inflection point.
2. The interconnect substrate as claimed in claim 1, wherein the first angle is from 10 degrees to 40 degrees, the second angle is from 120 degrees to 160 degrees, and the third angle is from 120 degrees to 160 degrees.
3. The interconnect substrate as claimed in claim 1, wherein, in the cross section, a first region of the insulating layer extending to a depth of 2 μm or less from a portion, facing the recess, of the second surface includes a second region including an inner wall surface of the via hole and a third region connected to the second region and positioned between the second region and the first point in plan view, and wherein a density of a filler in the second region is higher than a density of the filler in the third region.
4. The interconnect substrate as claimed in claim 1, wherein the second interconnect layer includes a first metal layer and a second metal layer, the first metal layer covering the surface of the recess, a portion of the second surface of the insulating layer, and an inner wall surface of the via hole, the second metal layer being provided on the first metal layer to fill the recess and the via hole.
5. The interconnect substrate as claimed in claim 1, wherein the via hole has an inverted truncated conical shape with a diameter thereof decreasing from an upper surface of the insulating layer toward the second surface.
6. The interconnect substrate as claimed in claim 3, wherein the third region of the insulating layer includes, at the second surface, an adhesion enhancing film in contact with the first surface of the first interconnect layer.
7. The interconnect substrate as claimed in claim 6, wherein the second surface of the insulating layer in the second region is free of any adhesion enhancing film.